InfRS: Incremental Few-Shot Object Detection in Remote Sensing Images

Wuzhou Li, Jiawei Zhou, Xiang Li, *Life Member, IEEE*, Yi Cao, Guang Jin, and Xuemin Zhang

***Abstract*— Few-shot detection in remote sensing images has witnessed significant advancements recently. Despite these progresses, the capacity for continuous conceptual learning still poses a significant challenge to existing methodologies. In this article, we explore the intricate task of incremental few-shot object detection (iFSOD) in remote sensing images. We present a pioneering transfer-learning-based technique, termed InfRS, designed to enable the incremental learning of novel classes using a restricted set of examples, while simultaneously preserving the knowledge learned from previously seen classes without the need to revisit old data. Specifically, we pretrain the detector using sufficient data from base datasets and then generate a set of classwise prototypes that represent the intrinsic characteristics of the data. In the incremental learning session, we design a hybrid prototypical contrastive (HPC) encoding module for learning discriminative representations. Furthermore, we develop a prototypical calibration strategy based on the Wasserstein distance to overcome the catastrophic forgetting problem. Comprehensive evaluations conducted with two aerial imagery datasets show that our InfRS effectively addresses the iFSOD issue in remote sensing imagery. Code is available at https://github.com/lyanna4869/InfRS.git.**

***Index Terms*— Incremental few-shot object detection (iFSOD), prototypical contrastive learning, remote sensing images.**

I. Introduction

OBJECT detection stands as a crucial and prominent task in remote sensing image analysis. In recent years, deep convolutional neural networks have greatly advanced object detection tasks [1], [2], [3]. Despite these advancements, such approaches often rely heavily on large-scale training datasets, which are costly and time-consuming to acquire. To tackle this issue, the emerging approach of few-shot object detection (FSOD) promises to reduce the need for extensive labeled datasets [4], [5], [6]. However, traditional FSOD methods typically require revisiting data from old (base) categories when expanding to new (novel) categories to achieve decent performance. This dependency introduces two main challenges: first, storing base training data from previous tasks raises concerns about computational costs and privacy issues; second, as Earth observation technologies advance and interests expand, new categories continuously emerge. Traditional FSOD approaches lack the flexibility and scalability required to enroll new categories incrementally. In contrast, humans can quickly grasp new concepts from just a few examples without the need to revisit previously acquired knowledge. Thus, it is highly desirable for machine learning models to gain the capability to learn incrementally in practical applications. Driven by these motivations, we study the more challenging task of incremental few-shot object detection (iFSOD) in remote sensing imagery in this study.

Ideally, an iFSOD model should achieve two goals: 1) efficiently learn novel categories from a limited number of annotated samples in a continual manner and 2) maintain performance on previously existing categories without access to old data. This presents challenges on two fronts: first, data scarcity can lead to overfitting in novel categories and second, the catastrophic forgetting issue may result in a drastic performance drop in base classes when expanding to novel categories. Since data from previous tasks are no longer accessible when learning novel categories, the problem of forgetting is further exacerbated. Prior iFSOD works [7], [8], [9] use a meta-learning strategy: the detector is initially pretrained on base categories and meta-trained with few-shot episodes to gain the ability to enroll new categories in the subsequent meta-testing stage. Recently, inspired by the prevalent two-stage fine-tuning approach (TFA), several works [10], [11] pretrain the base detector and transfer the model on novel data. Nevertheless, the existing iFSOD methods that are primarily developed for natural scene images tend to overlook the unique challenges presented by remote sensing imagery. Objects in remote sensing images typically have more complex backgrounds, diverse characteristics, and arbitrary orientations compared with those in natural scenes [2], [12]. These conditions cause insufficient discriminative feature learning, especially in a limited data scenario, rendering the models more vulnerable to the challenges of overfitting and catastrophic forgetting.

To tackle these challenges, we argue that the key to iFSOD lies in sufficient discriminative feature learning. Contrastive learning is a powerful approach for learning discriminative

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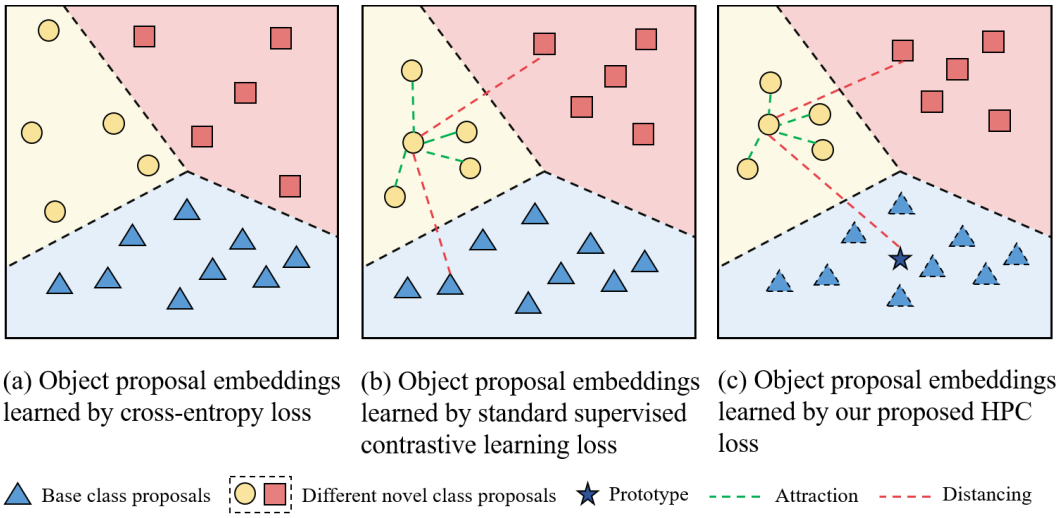


Fig. 1. Comparisons between object proposal embeddings learned through different paradigms. (a) Object proposal embeddings learned by a general softmax classifier, which are not discriminative enough. (b) Standard supervised contrastive learning, which depends on instancewise contrast to learn a discriminative feature distribution. (c) We propose an HPC encoding module that combines the advantages of both base class prototypes and novel instances. This provides explicit supervision to facilitate iFSOD.

features in various tasks [13], [14]. The fundamental idea behind contrastive learning is to draw similar samples (positive pairs) closer in the latent space, while separating dissimilar ones (negative pairs). However, base class samples are not accessible in the incremental learning stage, which makes conventional contrastive learning impractical. To address this issue, we propose to learn prototypes for base categories as representatives. As illustrated in Fig. 1, rather than explicitly sampling positive and negative instances, we use classwise prototypes as anchors for determining positive and negative pairs. Specifically, novel samples with the same labels are pulled together and pushed away from prototypes of base categories and all other novel categories. We introduce the HPC encoding module to leverage prototypical knowledge as reliable auxiliary supervision to foster more discriminative and robust representation learning for novel classes, while also preserving the intrinsic overall class distribution. Moreover, we propose a prototypical calibration strategy, incorporating prototypes and Wasserstein distance to apply additional regularization on the model's parameter update process, aiming to mitigate the catastrophic forgetting problem.

In summary, we propose InfRS, an incremental few-shot object detector in remote sensing imagery. To enhance representation learning for novel classes and to preserve existing knowledge, we introduce two novel components: the hybrid prototypical contrastive (HPC) encoding module and the prototypical calibration strategy. We summarize our contributions as follows.

1) Our approach enables learning of novel objects with limited examples incrementally and maintains detection performance on base classes without the need to revisit old data. To the best of our knowledge, this is the first study to tackle the iFSOD task in remote sensing images.
2) We introduce the HPC encoding module that captures the underlying structure and patterns within the data, thereby encouraging the model to learn discriminative and robust representations.
3) We develop a prototype calibration strategy which leverages the prototypical knowledge and Wasserstein distance to eliminate the need for restoring or replaying old data without compromising performance on base categories.
4) Extensive experiments with the NWPU VHR-10, DIOR and RSOD benchmark datasets showcase the efficacy of our InfRS for the iFSOD task in remote sensing images.

II. Related Work

A. Object Detection

Modern object detection models are typically grouped into two types: two-stage detectors [15], [16], [17], [18], [19], [20] and one-stage detectors [21], [22], [23], [24], [25]. Two-stage detectors initially extract candidate object proposals and then refine these proposals for precise classification and regression. In comparison, one-stage detectors bypass the proposal generation process and predict on feature maps in a single step. One-stage detectors generally have an edge in detection speed but may compromise in accuracy, especially for dense and small objects. Recently, transformer-based approaches such as DETR [26] and Deformable DETR [27] have shown promising performance, yet they still suffer from slow convergence issue. In this work, we opt for the classical two-stage detection architecture, Faster R-CNN, for iFSOD to secure a balanced compromise between performance and training efficiency.

For object detection tasks in remote sensing images, researchers have also made significant efforts. Several studies, such as R2-CNN [28], SSPNet [29], SCRDet [30], and SCRDet++ [31], focus on detecting objects of varying scales. RICNN [32] and [33] have developed rotation-invariant convolutional neural networks to handle objects with arbitrary orientations. More recent advancements, including RR-CNN [34], Polardet [35], S^2A-Net [36], RRPN [37], and [38], use oriented bounding box representations to enhance detection of oriented objects in aerial images.

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B. Few-Shot Object Detection

Prevailing FSOD models can be generally divided into two categories: meta-learning-based [39], [40], [41], [42], [43] and transfer-learning-based methods [44], [45], [46], [47], [48], [49].

Meta-learning-based methods obtain a meta-learner by constructing and solving a series of subtasks, and then leverage the task-level knowledge for quick adaptation to novel classes. Kang et al. [39] first propose FSRW, a meta-detector with a feature reweighting module that uses channelwise attention to aggregate query and support features. Meta R-CNN [40] introduces an instance-level attention mechanism, focusing on each region of interest (ROI) features instead of the entire feature map. FSIW [41] builds on FSRW by developing a more complex feature aggregation module and training on a balanced dataset. MetaDet [42] distinguishes itself by separating category-agnostic and category-specific parameter learning, facilitating the detection of novel classes. Meta-DETR [43] integrates meta-learning into the DETR framework to make predictions at the image level, moving away from region-based features.

Transfer-learning-based methods start with the pioneering TFA [44], which pretrains a base detector and then fine-tunes only the last layers of classification and regression for new data. Notably, TFA outperforms more complex meta-detectors. Building on this framework, MPSR [45] enhances TFA by learning to detect multiscale objects. FSCE [46] introduces learning contrastive-aware representations to mitigate misclassification issues. Retentive R-CNN [47] retains the RoI branch from the base detector, applying it as regularization during the fine-tuning stage to prevent performance degradation on base classes. Qiao et al. [48] propose decoupling the training of classification and box regression to address the contradiction between these two tasks. FADI [49] leverages the semantics of novel and associated base categories to enhance interclass separability by disentangling the classifiers for novel and base categories.

C. Few-Shot Object Detection in Remote Sensing Images

In the field of remote sensing, few-shot learning has primarily attracted research interest in image-level classification [50], [51], [52], [53], whereas fewer works have investigated the more challenging task of object detection. The pioneering work FSODM [54] introduces a meta-detector equipped with a feature pyramid network (FPN) module to extract feature maps with different scales from support samples, along with a feature reweighting module to effectively learn meta-information from these maps. Another meta-detector P-CNN [55] learns prototype representations from support images and then fuses the information to improve region proposal generation. Wang et al. [56] design a meta-learning-based method incorporating a diversity measurement module to enhance the exploitation of meta-knowledge from various meta-tasks. Subsequently, Wang et al. [57] propose a novel approach aimed at addressing the issues of feature misalignment and attention bias, which impede the performance of meta-learning-based few-shot detectors. For transfer-learning-based methods, Zhao et al. [58] propose the Pams-det which introduces path aggregation module to extract multiscale features for remote sensing objects. Zhang et al. [59] design a metric-learning-based loss to learn a more discriminative classifier and a representation compensation module to maintain performance on base classes.

To mitigate the loss of knowledge from the pretraining stage, DH-FSDet [60] integrates an additional prediction branch for fine-tuning novel classes. Recently, Zhang et al. [61] propose ST-FSOD to solve the incompletely annotated instances issue existed in the few-shot learning process. However, most FSOD studies focus on detecting novel instance in a nonincremental manner. In this study, we seek to bridge this gap for remote sensing imagery.

D. Incremental Few-Shot Object Detection

Motivated by the advancements and limitations of FSOD, Pérez-Rúa et al. [7] first explore the iFSOD task with the model ONCE. ONCE introduces a class-specific code generator through meta-learning to recognize novel classes. Similarly, Sylph [8] uses a hypernetwork, echoing ONCE's approach but decoupling localization from object classification. Feng et al. [10] propose a novel method that attaches a new classification header for each emerging novel class, alongside a bipath multiclass head to facilitate knowledge transfer. Rather than modifying the multiclass classifier, Zhang et al. [11] leverage responses of novel features and reorganizing base weights, aiming to transfer knowledge to novel classes that are relevant to base classes. Unlike prior studies, we propose a new architecture for iFSOD in remote sensing imagery, marking the first exploration of this task in this domain.

E. Contrastive Learning With Prototypes

Contrastive learning, also known as "learning by comparison," has proven effective in learning discriminative representations. Contrastive objectives are formulated to minimize distances between similar (positive) sample pairs and maximize distances between dissimilar (negative) pairs simultaneously in the embedding space. In self-supervised settings, positive pairs are obtained from differently augmented views of the same instance. In supervised settings, positive or negative pairs are determined with respect to an anchor instance depending on the class labels. For example, Li et al. [62] introduce prototypical contrastive learning in an unsupervised setting, generating prototypes as data cluster representatives to encourage instances to project closer to their corresponding prototypes. Otsuki et al. [63] extend this concept to multimodal language understanding tasks, learning data cluster prototypes from various domains to enable contrastive learning. Ouyang et al. [64] propose PCLDet to learn contrastive-aware features for fine-grained object detection.

F. Wasserstein Distance

It is often advantageous to establish a quantitative metric to measure the dissimilarity between two distributions.

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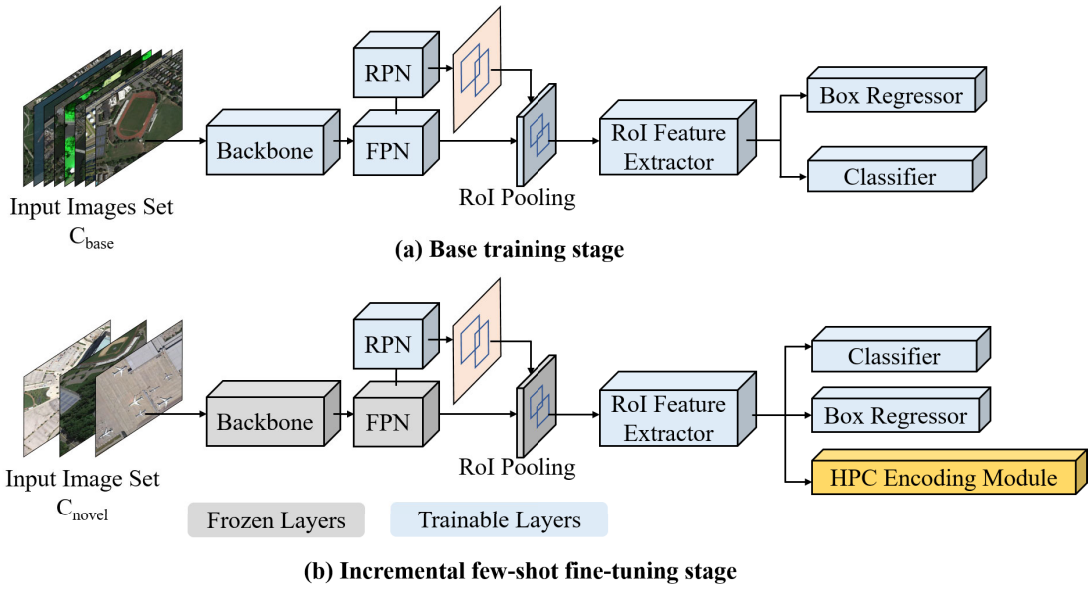


Fig. 2. Overall framework of our InfRS. (a) Initially, the model is pretrained on base data, from which we extract prototypes as representatives for each base class. (b) During the fine-tuning stage, the few-shot novel data are incrementally enrolled. The HPC encoding module leverages prototypical knowledge to learn proposal embeddings with interclass distinction and intraclass compactness. Furthermore, the prototypical calibration strategy, based on the Wasserstein distance, is introduced to ease the catastrophic forgetting effect.

For example, label smoothing [65], confidence penalty [66], and knowledge distillation [67] use information-based f-divergence to measure the difference in distributions. Despite its popularity, f-divergence focuses solely on the semantic relationships within the same class across two distributions and overlooks the structural information among difference classes. This is because it calculates each index of the output dimensions independently [68]. The Wasserstein distance [69] is defined as the minimum cost required to convert one distribution into the other. Wang et al. [70] introduce a Wasserstein-distance-based metric to replace intersection-over-union (IoU) in detecting tiny objects. Rubner et al. [71] adopt Wasserstein distance for image retrieval, as it allows for measuring the distance between partial matched distributions.

In light of this, we use the Wasserstein distance as a more suitable measure for computing distribution distances in iFSOD. As the incremental detector encounters an increasing number of novel classes, the Wasserstein distance can naturally be applied to compare its predicted distribution with its subset, which is predicted by the base detector and has partially matched indices.

III. Methodology

First, we introduce the problem setting for the iFSOD task. Subsequently, the framework for our InfRS is presented. Finally, the HPC encoding module and the prototypical calibration strategy are delineated.

A. Problem Definition

In iFSOD, our goal is to develop a model capable of incrementally learning to detect novel categories using limited annotated examples, without revisiting or forgetting previously learned base classes. We use two disjoint datasets: C_{base}, which comprises abundant annotated data for base classes, and C_{novel}, which contains limited training data for each novel class. Initially, the model is pretrained on C_{base} to establish a robust base detector. Subsequently, the base detector is expanded to include each novel class in C_{novel} incrementally through a fine-tuning process, without revisiting any data from C_{base}. The iFSOD detector is expected to be capable of effectively detecting all classes. Therefore, during the testing phase, we evaluate the overall performance using $C_{\text{all}} = C_{\text{base}} \cup C_{\text{novel}}$.

B. Method Overview

Fig. 2 depicts the overall architecture of our InfRS. Inspired by the prevalent FSOD approach TFA, we use a pretrain–fine-tuning scheme to achieve iFSOD in remote sensing images. We use the well-established Faster R-CNN with FPN as the base detector. In the Faster R-CNN detection pipeline, the backbone network processes input images to generate feature maps with potential region proposals predicted by the regional proposal network (RPN). Proposals are then pooled into fixed-size feature maps and converted into RoI feature vectors by the RoI feature extractor, which consists of two fully connected (FC) layers. Finally, the classifier and regressor determine the categories and bounding box coordinates of any detected foreground objects.

After the base training stage, we extract the RoI features and compute the mean vector of RoI features for each class, using it as the prototype representation of that class. We also generate a corresponding pseudo classification probability distribution for each prototype. In the incremental learning stage, we freeze the parameters of the class-agnostic backbone along with the FPN and fine-tune the model on C_{novel}. To guide the model to build more discriminative features, we propose the HPC encoding module and leverage the base class prototypes in a contrastive manner. Moreover, we introduce the prototypical calibration strategy, enabling the model to maintain the detection performance for base classes.

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C. Prototype Generation

We aim to learn robust prototypes containing high-level discriminative information for each base class. The prototype generation procedures are formulated as follows. We interpret the network function of base detector as a trainable feature learner f_θ parameterized by θ, followed by the final classification and regression layers. Consider a dataset of N labeled examples: $X = \{(x_1, y_1), \ldots, (x_N, y_N)\}$, where each x_i represents a D-dimensional feature vector from an instance in $\mathbb{R}^D$, and y_i denotes its label, which ranges from 1 to K. X_k is a subset which contains N_k instances with the class k in X. The RoI feature of each instance is computed through $f_\theta : \mathbb{R}^D \to \mathbb{R}^M$. Typically, $M = 1024$ in Faster R-CNN framework. Based on these RoI features, we represent the class k with its prototype representation p_k

$$p_k = \frac{1}{|N_k|} \sum_{(x_i, y_i) \in X_k} f_\theta(x_i). \tag{1}$$

Assume we take the prototypes as the activation at the penultimate layer, and w_i as the weight parameters of the softmax layer, then the input to the softmax layer o can be written as

$$o = \sum_{i=1}^{M} w_i \, p_{ki}. \tag{2}$$

In the above equation, p_{ki} represents the ith component of the prototype p_k, and the predicted discrete probability distribution of prototype p_k across K classes can be formulated as

$$d_k = \frac{\exp(o_k)}{\sum_{j=1}^{K} \exp(o_j)}. \tag{3}$$

D. HPC Encoding Module

General supervised contrastive learning leverages label information to determine positive samples and negative ones in relation to the anchor. Therefore, it allows supervised contrastive learning to contrast the set of all the samples with the same label as positives against the negatives from the rest of the batch for good performance [13], [46]. However, since samples from C_{base} are not available during the incremental learning stage, applying contrastive learning only to the limited samples of novel classes can lead to suboptimal performance. In light of this, we use the prototypical knowledge extracted from base data to provide the explicit supervision information in supervised contrastive learning, allowing the model to learn more powerful representations for novel classes, thus boosting the overall performance of iFSOD.

To integrate prototypical contrastive learning into the object detection framework, we design an FC layer with dimensions 1×128 to encode the RoI features and class prototypes, as shown in Fig. 3. The similarity between the encoded vectors is measured, and a contrastive objective is optimized to draw vectors with the same labels closer while separating those with differing labels in the latent space. Specifically, for a novel instance, we learn a representation that increases dissimilarity with the prototypes of base classes and novel samples from different categories, while minimizing the distance between samples belonging to the same class.

Our HPC loss function adopts a structure similar to that used in FSOD [46] and classification [13], with considerations for leveraging prototypical knowledge in iFSOD. We denote $\{z_i, u_i, y_i\}_{i=1}^{N}$ as the set of N RoI features in a mini-batch, where z_i represents the vector embedded by the HPC encoding module, u_i is the IoU score matching with its ground-truth box, and y_i is the label. Let $p = \{p_1, p_2, \ldots, p_K\}$ be a set of class prototypes projected by the HPC encoding module, with K denoting the total number of classes. We define the similarity score as $\text{sim}(z_i, z_j) = (z_i \cdot z_j / |z_i||z_j|)$, and the HPC loss function is formulated as (4) and (5), shown at the bottom of the next page, where N_{y_i} is the total number of RoI features with the same label, τ is a scalar temperature hyperparameter as in [13], and $\mathbb{I}$ is an indicator function. $\mathbb{I}\{u_i \geqslant \phi\}$ represents an IoU threshold. We empirically set $\phi = 0.7$ so that the most centrally RoI features are sampled to train the HPC encoding module. $\mathbb{I}(y_i = y_j) = 1$ if the ith RoI feature and the jth one belong to the same class, and 0 otherwise. Optimizing this loss function increases the distance between novel instances and base class prototypes, while simultaneously bringing instances of the same category closer in the embedding space. This results in tighter data clusters for each category, thus deriving well-separated decision boundaries and reducing misclassification on overall classes.

E. Prototypical Calibration Strategy

The data scarcity setting in iFSOD not only leads to severe overfitting for novel classes but also results in catastrophic performance degradation on base classes. In this section, we present a prototypical calibration strategy to mitigate the catastrophic forgetting problem in the incremental learning phase. Specifically, we capitalize on the prototypical knowledge acquired during the base training stage to provide auxiliary prior information during the adaptation process for novel classes. We begin by extracting the prototype and preserving its output distribution as predicted by the base detector, as shown in Fig. 3. We then encourage the distribution predicted by the fine-tuned model on the prototype approximating its original distribution.

Previous works [47], [65], [66], [67] use traditional information divergence-based loss functions to measure the difference between two distributions. However, these divergences only consider the same index over the two distributions and treat each dimension independently, ignoring any metric structure between different classes and potentially resulting in inaccurate or unstable measurements when the distributions are partially overlapping. This is particularly problematic when our model expands to novel classes. In contrast, the Wasserstein distance provides a robust and intuitive notion of dissimilarity for probability measures by considering the minimum "cost" required to transform one distribution into another. This approach can naturally operate on distributions with partially matched indices, and variable-length representations of the distributions, ensuring more accurate and stable results. In light of this, we adopt the Wasserstein distance

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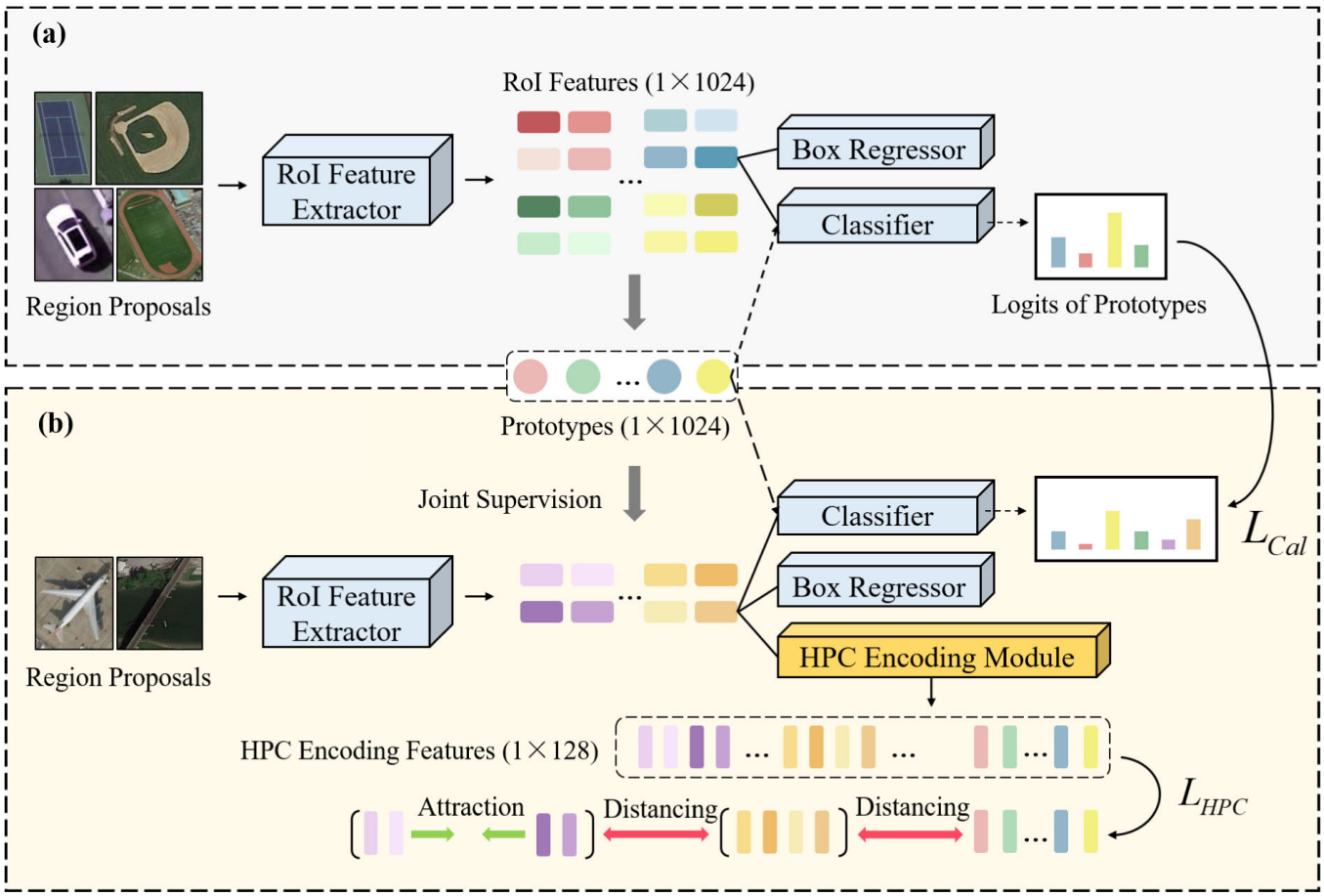


Fig. 3. Illustration of our proposed HPC encoding module and the prototypical calibration strategy. We extract the prototypes of base classes to provide extra supervision for discriminative feature learning and to generate classification logits for regularization in the incremental learning stage. (a) Base training stage. (b) Incremental few-shot fine-tuning stage.

as an alternative to f-divergences for measuring distribution dissimilarity in iFSOD.

Concretely, consider two sets of distributions over a prototype denoted by p_k. Let $D = \{d_j \mid j = 1, 2, \ldots, K\}$ represent the set predicted by the base detector, where K is the total number of classes in the base model. Similarly, let $F = \{f_i | i = 1, 2, \ldots, Q\}$ represent the set predicted by the fine-tuned model, where Q is the total number of classes in the fine-tuned model. Here, f_i denotes the classification probability corresponding to class index i, and similarly, d_j for class index j.

To calculate the Wasserstein distance between the two distributions, one must solve the transportation problem outlined below

$$\begin{aligned} \underset{T}{\text{minimize}} \quad & \sum_{i=1}^{Q}\sum_{j=1}^{K} C_{ij}T_{ij} \\ \text{subject to} \quad & T_{ij} \geq 0, \quad i = 1, \ldots, Q, \quad j = 1, \ldots, K \\ & \sum_{j=1}^{K} T_{ij} = f_i, \quad i = 1, \ldots, Q \\ & \sum_{i=1}^{Q} T_{ij} = d_j, \quad j = 1, \ldots, K. \end{aligned} \tag{6}$$

$$\mathcal{L}_{\text{HPC}} = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}\{u_i \geqslant \phi\} \cdot L_{z_i} \tag{4}$$

$$\begin{aligned} L_{z_i} = & \frac{-1}{N_{y_i} - 1} \sum_{j=1, j\neq i}^{N} \mathbb{I}(y_i = y_j) \\ & \cdot \log\left(\frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\left(\sum_{l=1, l\neq i}^{N} \exp(\text{sim}(z_i, z_l)/\tau) + \sum_{k=1}^{K} \exp(\text{sim}(z_i, p_k)/\tau)\right)}\right) \end{aligned} \tag{5}$$

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In this equation, $T \in \mathbb{R}^{Q\times K}$ represents the transportation matrix and $C \in \mathbb{R}^{Q\times K}$ the cost matrix. C_{ij} indicates the per-unit cost of transporting probability from category i to category j. T_{ij} represents the volume of probability transferred from category i to category j. We use the Sinkhorn algorithm [72], [73], [74] to compute the solution $\tilde{T}$ to (6), which can solve the optimal transportation problem with a linear convergence rate by imposing a straightforward structure on the transportation matrix.

For K prototypes representing the total K base classes, we introduce a plug-and-play regularization term denoted as L_{cal}, defined as

$$W_k = \sum_{i=1}^{Q}\sum_{j=1}^{K} \tilde{T}_{ij} \cdot C_{ij} \tag{7}$$

$$L_{\text{cal}} = \frac{1}{K}\sum_{k=1}^{K} W_k. \tag{8}$$

By optimizing L_{cal}, we ensure the fine-tuned model predicts probability distributions similar to those of the base detector for the same prototype, thus preserving the knowledge acquired during the pretraining stage.

F. Training Objectives

The pretraining stage uses Faster R-CNN loss function: $L_{\text{base}} = L_{\text{cls}} + L_{\text{reg}} + L_{\text{rpn}}$, where L_{cls} and L_{reg} are the outputs of the last object classifier and regressor, respectively, and L_{rpn} is a cross-entropy loss for generating object region proposals. When expanding to novel classes in the incremental learning stage, the contrastive loss L_{hpc} and the calibration loss L_{cal} are jointly optimized with L_{base}. To sum up, our loss function is defined as

$$L = L_{\text{cls}} + L_{\text{reg}} + L_{\text{rpn}} + \lambda_1 L_{\text{hpc}} + \lambda_2 L_{\text{cal}}. \tag{9}$$

In the above formula, λ_1 and λ_2 are two factors designed to balance the importance of losses.

IV. Experiments

In this section, we begin by describing the datasets and outlining the implementation details. Subsequently, we present comprehensive results compared with the baseline model and state-of-the-art method. We maintain consistency with iFSOD data settings and evaluation protocols to ensure a fair comparison. Finally, we conduct ablative studies to validate the effectiveness of the proposed components.

A. Datasets

1) NWPU VHR-10: This dataset comprises 650 images distributed in ten geospatial object classes for detection: airplane, vehicle, ship, harbor, baseball diamond, basketball court, tennis court, ground track field, storage tank, and bridge. The spatial resolutions of the images in this dataset vary between 0.5 and 2 m.

2) DIOR: This dataset contains 23 463 aerial images with 192 472 object instances distributed into 20 categories: airplane, wind mill, airport, train station, baseball field, tennis court, bridge, storage tank, chimney, ship, dam, expressway service area, stadium, expressway toll station, harbor, golf course, vehicle, ground track field, overpass, and basketball court. The spatial resolution varies between 0.5 and 30 m, with each image measuring 800×800 pixels in size. The training, validation, and testing sets contain 5862, 5863, and 11 738 images, respectively.

3) RSOD: This is a four-class object detection dataset which contains 976 remote sensing images with 6950 instances. The spatial resolution of these images ranges from 0.3 to 3 m. The four classes are aircraft, oil tank, overpass, and playground.

To establish the data setup for iFSOD, we follow previous FSOD studies, except that the base data are only available in the pretraining stage and the novel categories are learned sequentially. In the NWPU VHR-10 dataset, the ten classes are distributed among three novel categories and seven base categories. We partition the dataset into two subsets: 70% of the images (454 in total) are allocated for training, while the rest 30% are designated for testing. Similarly, within the DIOR dataset, 15 categories are designated as novel, while the remaining five are considered base categories. The model is trained on the training set and its performance is evaluated on the validation set. For the RSOD dataset, we randomly select one category as novel and the other three as base categories, with 488 images used for training and the remaining ones for testing. For each dataset, we use three different splits to perform iFSOD. In the incremental learning stage, only K-shot instances from each novel category are randomly chosen for training. We set $K = 1, 3, 5, 10$ for the NWPU VHR-10 and RSOD datasets, and $K = 3, 5, 10, 20$ for the DIOR dataset to account for their larger scale. To quantitatively evaluate the performance, the mean average precision (mAP) at an IoU threshold of 0.5 is reported. We run inference on the overall datasets which include both base and novel categories and report the performance of novel categories (nAP50), base categories (bAP50), and overall categories (mAP50) separately.

B. Implementation Details

We use the prevalent FSOD method TFA as the baseline model and adopt ResNet-101 with FPN as backbone network for all the experiments. We use feature pyramids from P_2 to P_6. No data augmentation of any kind is applied in our analysis. The SGD optimizer is adopted, with a momentum of 0.9 and a weight decay set at 0.0001. Models are warmed up for the first 100 iterations. For the NWPU VHR-10 and RSOD datasets, we pretrain the models with a batchsize of 4 for 24 epochs and fine-tune till full convergence. The initial learning rate is 0.001 and decreases tenfold at epochs 12 and 18 in the pretraining stage. During the incremental learning stage, the learning rate is 0.0001 and decreases tenfold at 4000 iterations. For the DIOR dataset, we set the batchsize at 8 and pretrain the models for 18 epochs and fine-tune till full convergence. The learning rate is 0.005 and decreases tenfold at epochs

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TABLE I
IFSOD PERFORMANCE ON NOVEL CATEGORIES OF THE NWPU VHR-10 DATASET

Method	Baseline TFA				FSCE				RS-DMK				InfRS			
Novel Set 1	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot
airplane	0.11	0.41	0.51	0.62	0.18	**0.43**	**0.52**	0.70	0.15	0.27	0.42	0.50	**0.23**	0.41	0.51	**0.72**
baseball diamond	0.18	0.42	0.67	0.85	0.24	**0.56**	0.72	0.68	**0.33**	0.53	**0.76**	0.83	0.19	0.50	0.70	**0.87**
tennis court	0.21	0.38	0.44	0.60	0.49	0.48	0.54	**0.67**	0.47	0.55	0.53	0.66	**0.54**	**0.59**	**0.57**	0.63
mean	0.17	0.40	0.54	0.64	0.30	0.49	0.59	0.74	0.32	0.45	0.57	0.66	**0.32**	**0.50**	**0.60**	**0.74**
Novel Set 2	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot
ship	0.09	0.43	0.54	0.66	0.09	0.57	0.63	0.70	0.09	**0.58**	0.59	0.67	**0.09**	0.51	**0.64**	**0.73**
storage tank	0.45	**0.82**	0.72	0.74	0.59	0.72	0.75	0.77	**0.62**	0.77	0.76	0.77	0.51	0.81	**0.77**	**0.82**
vehicle	0.09	0.14	0.15	0.13	0.02	0.14	0.15	0.23	0.04	0.09	0.17	0.19	**0.09**	**0.14**	**0.18**	**0.23**
mean	0.21	0.46	0.47	0.51	0.23	0.48	0.51	0.57	**0.25**	0.48	0.51	0.54	0.23	**0.49**	**0.53**	**0.59**
Novel Set 3	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot
basketball court	0.09	0.21	0.27	0.35	**0.30**	0.19	0.31	0.50	0.27	0.14	0.18	0.53	0.24	**0.28**	**0.37**	**0.54**
ground track field	**0.21**	0.46	0.44	0.51	0.19	**0.54**	0.51	0.58	0.18	0.53	**0.52**	0.56	0.19	0.50	0.42	**0.58**
harbor	0.17	0.03	0.09	0.27	0.08	0.02	0.03	0.24	0.09	**0.10**	**0.10**	0.16	**0.17**	0.02	0.07	**0.28**
mean	0.16	0.23	0.27	0.38	0.19	0.25	0.28	0.45	0.18	0.26	0.27	0.42	**0.20**	**0.27**	**0.29**	**0.47**

TABLE II
IFSOD PERFORMANCE ON BASE CATEGORIES OF THE NWPU VHR-10 DATASET OF SPLIT 1

Method	Baseline TFA				FSCE				RS-DMK				InfRS			
	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot
ship	0.90	0.91	0.91	**0.91**	0.89	0.91	0.91	0.89	0.90	0.90	0.91	0.90	**0.90**	**0.91**	**0.91**	0.90
storage tank	0.89	0.89	**0.90**	0.89	0.90	0.89	0.89	0.87	0.90	0.89	0.90	0.90	**0.95**	**0.89**	0.89	**0.90**
basketball court	0.98	0.61	0.43	0.61	**0.99**	**0.98**	0.95	0.92	0.98	0.95	0.94	0.76	0.97	0.97	**0.97**	**0.95**
ground track field	0.88	0.89	**0.91**	0.81	0.90	**0.91**	0.90	0.86	**0.91**	0.90	0.90	0.89	0.88	0.86	0.86	**0.89**
harbor	**0.88**	**0.88**	0.84	0.86	0.75	0.79	0.80	0.76	0.71	0.84	0.85	0.85	0.82	0.84	**0.88**	**0.87**
bridge	0.84	0.85	**0.88**	0.81	0.85	**0.86**	0.86	0.83	0.85	0.83	0.82	**0.83**	**0.88**	0.78	0.85	0.82
vehicle	0.88	0.88	**0.89**	0.89	0.89	**0.90**	0.88	0.86	0.90	0.90	0.89	**0.90**	**0.92**	0.87	0.87	0.89
mean	0.89	0.84	0.82	0.82	0.88	**0.89**	0.88	0.85	0.88	0.89	0.89	0.86	**0.90**	0.87	**0.89**	**0.89**

11 and 14 in the pretraining stage. During the incremental learning stage, the learning rate is 0.0001 and decreases tenfold at 8000 iterations. Regarding hyperparameters, τ is set to 0.1 in (5), and λ_1 and λ_2 are set to 0.5 in (9). We train and test the models using an NVIDIA GeForce RTX 3090Ti GPU of 24-GB memory.

C. Comparison of Incremental Few-Shot Detection Results

We compare our InfRS with the baseline model TFA [44], the instance-level contrastive learning-based method FSCE [46] designed for natural scene images, and the state-of-the-art FSOD method RS-DMK [75] on aerial images. All the methods adopt the same backbone network, and no additional data augmentation is used to ensure a fair comparison.

1) Results on NWPU VHR-10 Dataset: Tables I and II present the iFSOD results of our InfRS on the NWPU VHR-10 dataset for both novel and base categories. As shown in these tables, our InfRS consistently outperforms other methods in most scenarios. For example, in the ten-shot setting, our approach achieves mean mAPs of 74%, 59%, and 47% for novel classes across three different splits, representing improvements of 15.6%, 15.7%, and 23.7% over the baseline method, respectively. In the five-shot and ten-shot settings, our method achieves the highest performance across all three splits, demonstrating a clear advantage over FSCE and RS-DMK as more training samples become available. In extreme low-data scenarios, our InfRS also performs competitively, often surpassing other methods. For instance, in the one-shot setting for the tennis court class, our InfRS achieves a remarkable leap in performance (45%) compared with the baseline TFA, demonstrating its robustness and generalizability.

Regarding base classes, take split 1,for example, our model maintains the base class performance at 90%, 87%, 89%, and 89% in the $K = 1, 3, 5, 10$ shot settings, outperforming the baseline TFA by 1.1%, 3.6%, 8.5%, and 8.5%, respectively. As more novel samples are available, our InfRS demonstrates consistent and often superior performance on most base categories, with up to +0.04 bAP50 and +0.03 bAP50 above FSCE and RS-DMK in the ten-shot setting, respectively. Our model can effectively detect novel categories in incremental few-shot scenario and maintain the detection performance for base categories. We further present the qualitative results of our model in Fig. 4.

2) Results on DIOR Dataset: The detection performance of the baseline TFA and our InfRS on the novel categories of the

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TABLE III
IFSOD PERFORMANCE ON NOVEL CATEGORIES OF THE DIOR DATASET

Method	Baseline TFA				FSCE				RS-DMK				InfRS			
Novel Set 1	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot
airport	0.07	0.15	0.13	0.18	0.08	0.10	0.13	0.18	0.09	0.09	0.10	0.14	**0.10**	**0.16**	**0.15**	**0.21**
bridge	**0.09**	0.09	0.09	0.12	0.04	0.07	0.18	0.11	0.07	0.14	0.17	0.17	0.08	**0.15**	**0.18**	**0.18**
ground track field	0.24	**0.31**	0.27	0.35	0.26	0.28	0.23	0.30	0.25	0.26	0.20	0.27	**0.28**	0.27	**0.34**	**0.37**
vehicle	0.16	0.15	0.29	0.35	0.16	0.15	0.28	0.33	0.21	0.18	0.29	0.33	**0.22**	**0.20**	**0.35**	**0.35**
storage tank	0.57	0.67	0.67	0.67	**0.67**	0.68	0.64	0.66	0.61	0.64	0.63	0.62	0.66	**0.68**	**0.67**	**0.67**
mean	0.23	0.27	0.29	0.33	0.24	0.26	0.28	0.32	0.25	0.26	0.28	0.31	**0.27**	**0.29**	**0.34**	**0.35**
Novel Set 2	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot
airplane	0.32	0.56	0.58	0.67	**0.55**	0.48	0.48	0.74	0.32	0.45	0.67	0.70	0.48	**0.69**	**0.74**	**0.74**
baseball field	0.60	0.49	0.68	0.78	0.54	0.68	0.77	**0.82**	0.75	**0.76**	0.77	0.78	**0.78**	0.71	**0.81**	0.81
tennis court	0.65	0.60	0.70	**0.77**	0.60	0.69	0.70	0.77	0.73	0.64	0.75	0.75	**0.75**	**0.69**	**0.76**	0.76
train station	0.03	0.11	0.16	0.15	**0.12**	0.10	0.06	0.18	0.03	0.09	0.04	0.14	0.02	**0.11**	**0.16**	**0.19**
wind mill	0.15	**0.18**	0.19	0.27	0.15	0.15	0.15	0.24	0.13	0.18	0.16	0.21	**0.16**	0.16	**0.19**	**0.29**
mean	0.35	0.39	0.46	0.53	0.39	0.42	0.43	0.55	0.39	0.42	0.48	0.52	**0.44**	**0.47**	**0.53**	**0.56**
Novel Set 3	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot
expressway service area	0.08	**0.14**	**0.18**	0.21	0.07	0.08	0.13	**0.23**	0.11	0.05	0.15	0.16	**0.13**	0.10	0.12	0.20
harbor	0.03	0.07	0.07	0.12	0.09	0.09	**0.14**	0.16	**0.11**	0.06	0.14	0.15	0.03	**0.16**	0.12	**0.18**
overpass	0.10	**0.22**	0.25	0.30	0.12	0.22	0.22	0.21	0.16	0.22	0.22	0.29	**0.17**	0.21	**0.28**	**0.30**
ship	**0.09**	0.09	0.12	0.15	0.02	0.09	0.09	0.09	0.01	0.09	0.09	0.07	0.04	**0.26**	**0.24**	**0.27**
stadium	0.29	0.24	0.48	0.40	**0.50**	**0.50**	0.53	0.57	0.38	0.48	**0.58**	**0.56**	0.29	0.25	0.49	0.52
mean	0.13	0.15	0.22	0.23	**0.16**	0.20	0.22	0.25	0.15	0.18	0.24	0.25	0.13	**0.20**	**0.25**	**0.28**

TABLE IV
IFSOD PERFORMANCE ON BASE CATEGORIES OF THE DIOR DATASET OF SPLIT 1

Method	Baseline TFA				FSCE				RS-DMK				InfRS			
	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot	3-shot	5-shot	10-shot	20-shot
airplane	0.90	**0.91**	**0.91**	0.91	0.91	0.91	0.91	0.91	0.90	0.86	0.82	0.82	**0.91**	0.90	0.90	**0.91**
baseball field	0.79	0.90	**0.89**	**0.90**	0.90	0.89	0.88	0.85	0.87	0.81	0.62	0.79	**0.90**	**0.90**	0.81	0.89
basketball court	**0.80**	0.73	0.79	**0.80**	0.80	0.80	0.79	0.62	0.76	0.68	0.65	0.63	0.77	**0.77**	**0.79**	0.78
chimney	0.09	**0.82**	0.72	**0.81**	0.82	0.81	0.64	0.53	0.79	0.69	0.64	0.60	**0.82**	0.81	**0.81**	0.77
dam	0.33	0.09	0.17	0.16	0.32	0.31	0.25	0.21	0.38	0.27	0.24	0.21	0.30	**0.32**	**0.31**	**0.27**
expressway service area	**0.59**	0.53	0.53	0.53	0.59	0.58	0.56	0.59	**0.57**	0.56	0.56	0.56	0.57	**0.58**	**0.59**	**0.57**
expressway toll station	0.43	0.53	0.53	0.54	0.54	0.52	0.54	**0.60**	**0.56**	0.54	**0.56**	0.51	0.53	**0.53**	0.53	0.53
golf course	0.57	0.58	0.56	0.41	**0.61**	**0.62**	**0.59**	**0.65**	0.47	0.41	0.42	0.54	0.55	0.54	0.55	0.59
harbor	0.35	0.22	0.22	0.16	0.34	0.30	0.30	**0.35**	**0.37**	0.28	**0.31**	0.33	0.32	**0.30**	0.30	0.30
overpass	0.22	0.07	0.09	0.06	0.25	0.09	0.09	0.07	**0.40**	**0.24**	**0.27**	0.22	0.27	0.15	0.22	**0.22**
ship	**0.80**	0.72	0.72	0.72	0.80	**0.78**	**0.79**	**0.80**	0.71	0.76	0.70	0.70	0.71	0.72	**0.72**	**0.72**
stadium	0.52	0.54	0.27	0.36	0.81	0.80	0.63	0.62	**0.86**	**0.84**	0.75	0.73	0.80	0.76	**0.79**	**0.78**
tennis court	0.82	0.82	0.81	0.82	0.80	0.79	0.81	**0.83**	0.82	0.82	0.78	0.76	**0.82**	**0.82**	0.82	**0.82**
train station	**0.29**	0.20	0.27	0.29	0.17	0.23	**0.31**	**0.33**	0.23	**0.26**	0.23	0.26	0.14	0.25	0.28	0.28
wind mill	0.70	0.71	**0.71**	**0.71**	0.70	0.67	0.71	0.63	**0.72**	0.69	0.65	0.63	0.70	0.70	0.70	0.70
mean	0.55	0.56	0.55	0.54	0.62	0.60	0.59	0.57	**0.63**	0.58	0.55	0.55	0.61	**0.60**	**0.61**	**0.61**

DIOR dataset is presented in Table III. Although the DIOR dataset is quite challenging due to objects of various scales and complex structures, our model effectively learns to detect novel categories without forgetting base categories. Taking split 1 as an example, our model achieves mean mAPs of 27%, 29%, 34%, and 35% on novel categories with three, five, ten, and 20 examples, respectively, surpassing the baseline model by 17.4%, 7.4%, 17.2%, and 6.1%. For split 2, our model outperforms the current state-of-the-art by 12.8%, 11.9%, 12.5%, and 1.8% in the $K = 3, 5, 10, 20$ shot settings. For base classes, as the number of novel examples increases, our model effectively maintains performance, showing improvements of 10.9%, 7.1%, 10.9%, and 13.0% compared with the baseline under the same settings, as detailed in Table IV. Equipped with the HPC encoding module and the utilization of prototypical calibration strategy, our method demonstrates its superiority in most iFSOD scenarios.

3) Results on RSOD Dataset: The iFSOD results for the RSOD dataset are presented in Table V. Our InfRS model consistently outperforms the baseline models TFA, FSCE, and RS-DMK on novel categories in most scenarios across all splits. For instance, in split 2, InfRS achieves mAPs of 13%, 19%, 42%, and 61% in the $K = 1, 3, 5, 10$ shot settings for the novel category (oil tank), outperforming RS-DMK by 30.0%, 18.8%, 7.7%, and 29.8%, respectively. For base classes, our InfRS effectively maintains the performance at 92%, 92%, and 91% in split 2 under the $K = 3, 5, 10$ shot settings, surpassing the baseline TFA by 2.2%, 3.4%, and

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TABLE V
IFSOD PERFORMANCE ON ALL CATEGORIES OF THE RSOD DATASET

Method	Baseline TFA				FSCE				RS-DMK				InfRS			
Split 1	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot
aircraft (novel)	0.06	0.09	0.11	0.14	0.06	0.09	0.12	0.16	0.06	0.09	0.11	0.15	**0.07**	**0.09**	**0.12**	**0.18**
oil tank (base)	0.88	0.91	0.90	0.91	0.90	0.91	0.91	0.90	0.90	0.90	0.90	0.90	**0.90**	**0.91**	**0.91**	**0.91**
overpass (base)	0.82	0.81	0.83	0.80	0.81	0.77	0.82	0.81	0.82	**0.85**	0.82	0.81	**0.84**	0.83	**0.83**	**0.84**
playground (base)	0.92	0.93	0.92	0.90	0.93	**0.98**	0.92	0.94	0.93	0.92	0.95	0.91	**0.94**	0.97	**0.98**	**0.96**
mean	0.67	0.69	0.69	0.69	0.68	0.69	0.69	0.70	0.68	0.69	0.70	0.70	**0.69**	**0.70**	**0.71**	**0.72**
Split 2	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot
oil tank (novel)	0.11	0.12	0.18	0.28	0.09	0.16	0.36	0.42	0.10	0.16	0.39	0.47	**0.13**	**0.19**	**0.42**	**0.61**
aircraft (base)	0.91	0.90	0.90	0.88	0.90	**0.91**	0.89	0.90	0.90	0.88	0.89	0.87	**0.91**	0.90	**0.90**	**0.90**
overpass (base)	0.87	0.83	0.77	0.75	0.85	0.78	0.81	0.77	0.85	0.83	0.83	0.82	**0.87**	**0.87**	**0.87**	**0.85**
playground (base)	0.99	0.98	**0.99**	0.99	0.99	0.99	0.98	0.98	0.98	0.98	0.98	0.96	**0.99**	**0.99**	0.98	**0.99**
mean	0.72	0.71	0.71	0.73	0.71	0.71	0.76	0.77	0.71	0.72	0.78	0.79	**0.73**	**0.74**	**0.79**	**0.84**
Split 3	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot	1-shot	3-shot	5-shot	10-shot
overpass (novel)	0.10	0.13	0.25	0.30	0.11	0.13	0.23	0.34	0.10	0.18	0.32	0.36	**0.11**	**0.20**	**0.32**	**0.41**
aircraft (base)	0.90	0.90	0.90	0.90	0.90	0.90	0.81	0.90	0.90	0.81	0.81	0.90	**0.90**	**0.90**	**0.90**	**0.90**
oil tank (base)	0.90	0.91	0.82	0.82	0.91	0.82	0.81	0.85	0.88	0.82	0.82	0.84	**0.91**	**0.89**	**0.90**	**0.91**
playground (base)	0.98	0.98	**0.99**	0.97	0.98	0.98	0.99	0.96	**0.99**	0.98	0.98	0.95	0.98	**0.98**	0.97	**0.97**
mean	0.71	0.72	0.74	0.74	0.72	0.71	0.71	0.76	0.72	0.70	0.73	0.76	**0.73**	**0.74**	**0.77**	**0.80**

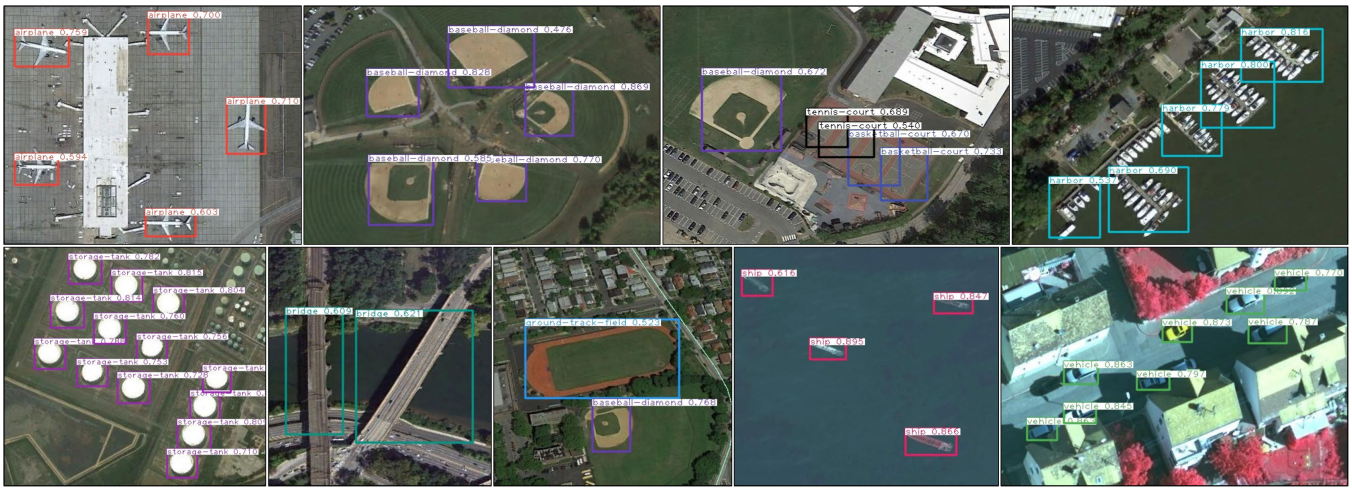


Fig. 4. Visualization of the ten-shot detection results using our InfRS on the NWPU VHR-10 dataset for split 1.

3.4%, respectively, further demonstrating the effectiveness of our proposed prototypical calibration strategy in mitigating the catastrophic forgetting effect.

D. Ablation Study

In this part, we conduct ablative experiments to carefully analyze the contribution of each component made to the ultimate performance of the InfRS model. All the ablation results are reported for split 1 of the NWPU VHR-10 dataset under a ten-shot scenario.

1) Effectiveness of the Proposed Two Components: We investigate the effectiveness and contribution of the HPC encoding module and the prototypical calibration strategy proposed in our approach. We conduct ablative experiments in a systematic, five-step progressive manner.

TABLE VI
ABLATIVE EXPERIMENTS ON THE HPC ENCODING MODULE AND THE PROTOTYPICAL CALIBRATION STRATEGY PROPOSED IN OUR InfRS

	fine-tune with novel classes	HPC loss	Calibration loss	bAP50	nAP50	mAP50
a	✗	–	–	0.91	0.0	0.64
b	✓	–	–	0.82	0.64	0.77
c	✓	✓	✗	0.85	0.74	0.82
d	✓	✗	✓	0.89	0.63	0.81
e (Ours)	✓	✓	✓	0.89	0.74	0.85

1) We pretrain the baseline model TFA using abundant data from base classes. Only the objective function of the Faster R-CNN framework is used.
2) We fine-tune the baseline model that has been trained on the base categories. The adaptation process uses the

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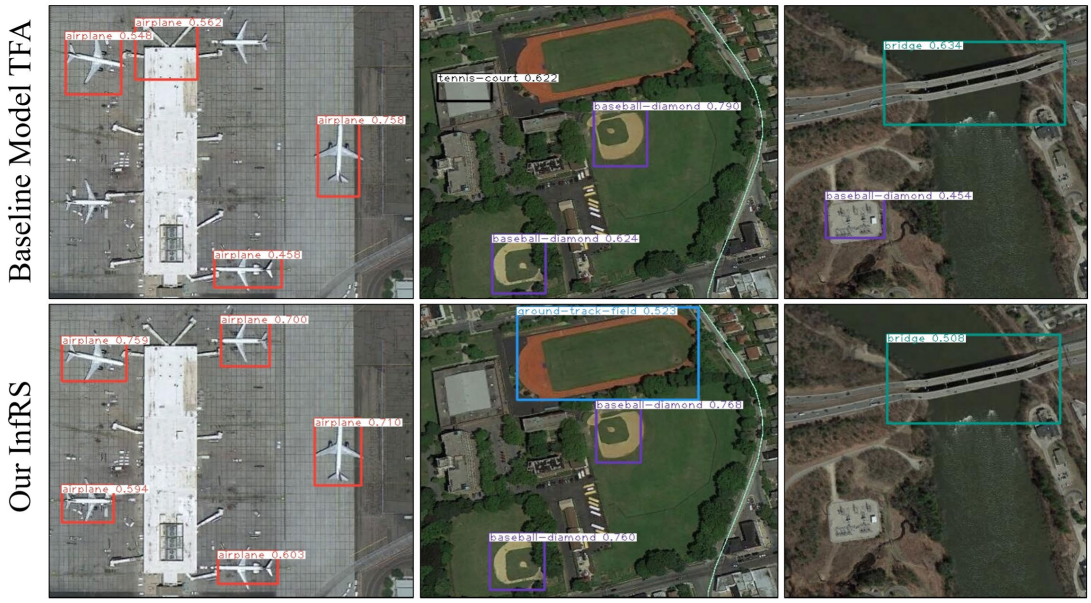
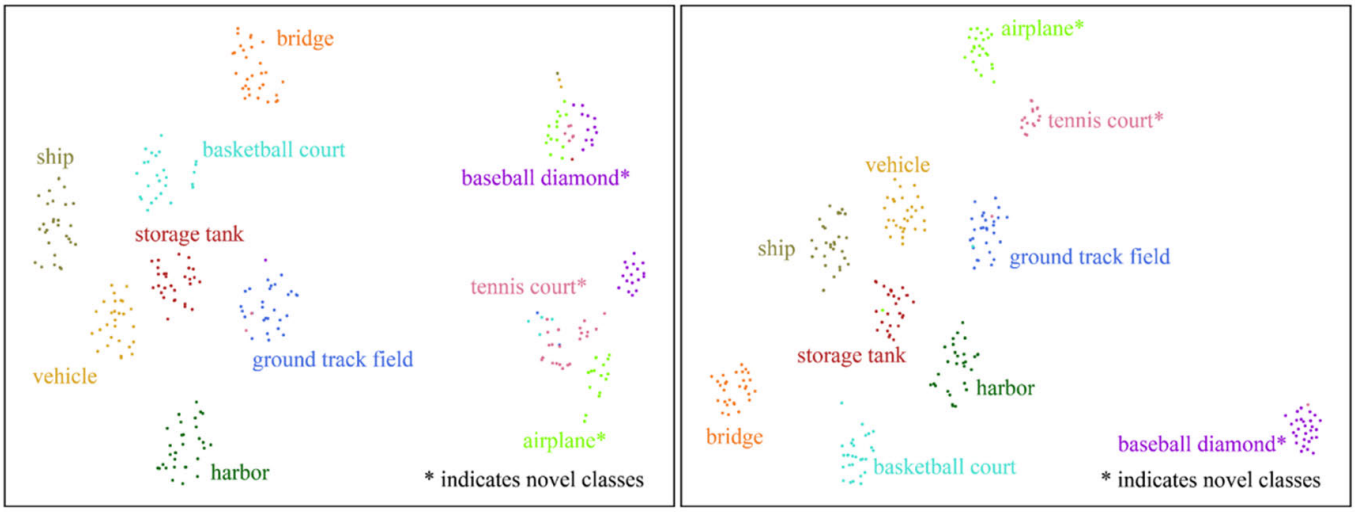


Fig. 5. Qualitative comparison of the baseline model TFA and our InfRS.



(a) Proposal embeddings learned by baseline model TFA (b) Proposal embeddings learned by our proposed InfRS

Fig. 6. Visualization of the proposal embeddings learned by the baseline model TFA and our InfRS using t-SNE: (a) overlap of the novel class clusters leads to suboptimal decision boundaries and overlaps with the base classes, notably between the basketball court and tennis court categories. (b) In contrast, the decision boundaries between classes are well-defined, showcasing clear separation among both base and novel classes.

objective function used in the Faster R-CNN framework exclusively, without any additional loss terms.

3) We introduce the HPC encoding module and further fine-tune the model on novel categories. In the incremental learning stage, we use both the objective function used in the Faster R-CNN framework and the HPC loss term.
4) We use the prototypical calibration strategy and fine-tune the base detector to learn novel categories. In this phase, we adopt the objective function used in the Faster R-CNN framework along with the calibration loss term.
5) We integrate both the HPC encoding module and the prototypical calibration strategy into the original model and fine-tune it for novel categories. In this stage, we jointly optimize both the HPC loss and the calibration loss proposed in our InfRS model, alongside the primary objective loss function of the Faster R-CNN framework.

The results of these ablative studies affirm the efficacy of the two components proposed in our InfRS model. As reported in Table VI **(a)**, the performance of the baseline TFA for novel classes is 0.0%, since novel data are not used in training. After adaptation for novel classes, the baseline model achieves an mAP of 64%, but the performance for base classes drops by 9.0% due to the catastrophic forgetting effect. To counter the overfitting problem, we introduce the HPC encoding module, which focuses on learning more discriminative features for novel classes. As the results of **(b)** and **(c)** in Table VI show, the performance reaches 74.0% for novel classes, marking a +10.0% increase in nAP50 over the baseline model, validating the effectiveness of the HPC encoding module. Furthermore, this module contributes to a notable improvement (+3.0%)

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for base classes, indicating that the enhanced classification accuracy for novel categories also benefits the learning of base categories. We conjecture that the base categories are also less prone to confusion with the novel categories, thereby further enhancing their accuracy. Analyzing the outcomes in **(b)** and **(d)**, the performance of base categories shows an enhancement of +4.0% in mAP50, highlighting the effectiveness of the prototypical calibration strategy in mitigating the catastrophic forgetting effect. Finally, equipped with the proposed two components, our InfRS achieves a satisfactory detection accuracy of 85.0% for overall classes as indicated in Table VI **(e)**. We further illustrate the comparative results of our InfRS against the baseline model TFA in Fig. 5.

2) Visualization and Analysis: To further validate the effectiveness of our InfRS, we visualize the embeddings of object proposals in the embedding space. The t-SNE [76] visualization of 30 samples per class learned from the NWPU VHR-10 dataset under a ten-shot scenario is presented in Fig. 6. We observe that our InfRS, powered by these two components, can learn well-defined boundaries between different classes.

V. Conclusion

This work investigates the challenging visual task of iFSOD in remote sensing imagery and presents a promising and novel solution named InfRS. We first formulate the iFSOD problem in remote sensing images and analyze the challenges associated with it. To address the overfitting issue, we propose the HPC encoding module to leverage both prototypical knowledge extracted from base categories and novel instances to learn contrastive-aware features. To address the catastrophic forgetting issue, we design the prototypical calibration strategy based on the Wasserstein distance to maintain the performance of base categories. Our InfRS effectively learns from data-scarce novel classes without revisiting or forgetting base classes and represents an initial foray into the domain of iFSOD in remote sensing.

Discussion: Despite the performance achieved by our proposed method InfRS, the need for updating the model with a few novel examples still limits its application in practical scenarios. In future work, we will continue to improve the model's scalability to achieve real-time and robust zero-shot object detection without compromising performance on base classes.

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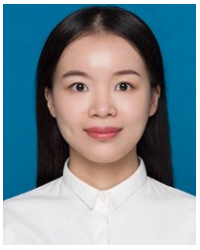
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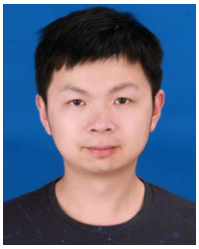
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